

1. STUDENT DETAILS

- * Name: Sonkalpa Borah
- * Roll No: CS-BTC24-08

2. OBJECTIVE

This assignment compares TCP communication performance between two modes: 'persistent' (single connection reused) and 'new_connection' (new TCP connection per message). The comparison is based on response time, throughput, and packet behavior observed in Wireshark.

3. MININET TOPOLOGY AND SETUP

- * Topology: 'h1 -- s1 -- h2'
- * Server host: 'h1' ('10.0.0.1')
- * Client host: 'h2' ('10.0.0.2')
- * Link profile: 'bw=5 Mbps', 'delay=50 ms'

Screenshots included:

- * 'screenshots/nodes.png'
- * 'screenshots/net.png'
- * 'screenshots/pingall.png'
- * 'screenshots/server_output.png'
- * 'screenshots/client_output.png'

4. PROGRAM DESIGN

MESSAGE FORMAT

- * Request: 'MSG_ID|MESSAGE_SIZE|MESSAGE_DATA'
- * Response: 'ACK|MSG_ID|RECEIVED_SIZE'

MODES

- * 'persistent': one connection for all 10 messages per message size
- * 'new_connection': one connection for every message

LOGGING AND OUTPUTS

- * 'server_log.txt': per-message receive + ACK log
- * 'message_response_log.csv': per-message response times (60 rows)
- * 'result_table.csv': summary metrics (6 rows)

5. EXPERIMENTAL RESULTS

Result table ('result_table.csv') contains 6 rows (2 modes x 3 sizes):

mode	message_size_bytes	average_response_time_seconds	throughput_bytes_per_second
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persistent	128	0.000114	1020271.77
persistent	512	0.000184	2627838.18
persistent	1024	0.000095	9715702.05
new_connection	128	0.000407	328750.70
new_connection	512	0.000442	1162581.58
new_connection	1024	0.000362	2818630.76

Observed trend:

- * Persistent mode has slightly lower response time than new-connection mode.
- * Throughput increases as message size increases.
- * New-connection mode incurs additional connection setup/teardown overhead.

6. GRAPHS

Generated graph files:

- * 'graphs/mode_vs_response_time.png'
- * 'graphs/message_size_vs_throughput.png'
- * 'graphs/message_response_time.png'

Graph interpretation:

- * 'mode_vs_response_time': persistent mode performs better on average.
- * 'message_size_vs_throughput': larger payload size yields higher throughput.
- * 'message_response_time': response-time variation is higher for new-connection due repeated handshakes.

7. WIRESHARK VERIFICATION

Display filter used: 'tcp.port == 5000'

Screenshots included:

- * 'screenshots/persistent_handshake.png'
- * 'screenshots/persistent_data_packets.png'
- * 'screenshots/persistent_connection_close.png'
- * 'screenshots/new_connection_multiple_handshakes.png'

Packet-level observations:

- * Persistent mode shows one initial three-way handshake and one final close.
- * New-connection mode shows repeated SYN/SYN-ACK/ACK and FIN sequences.

8. ANSWERS TO REQUIRED QUESTIONS

1. Why does TCP use a three-way handshake?

- It establishes synchronized sequence numbers and confirms both sides are ready to communicate reliably.

2. Which mode had lower response time?

- Persistent mode showed lower average response time.

3. Why does new-connection mode have more overhead?

- Every message requires handshake and teardown packets, adding delay and control traffic.

4. Which mode is better for CampusChat and why?

- Persistent mode, because chat is interactive and benefits from avoiding repeated connect overhead.

5. Which mode is suitable for NetAttend and why?

- New-connection mode can be suitable for short periodic updates where each request is independent and server state can stay simple.

6. How does message size affect throughput?

- Throughput increases with message size because protocol overhead is amortized over larger payloads.

7. How did Wireshark help verify TCP behavior?

- It visually confirmed handshake patterns, data exchanges, and connection close behavior for each mode.

9. CONCLUSION

The TCP implementation meets assignment requirements for both modes, produces the required CSV files and graphs, and Wireshark evidence confirms the expected protocol behavior differences between persistent and new-connection designs.